Assignment 1:

1. Create Android Project

- Open **Android Studio**
- New Project → **Empty Activity**
- App Name: `StudentDetails`
- Language: **Java**
- Finish

---

2. Design the Layout (Proper Alignment)

We'll use a **GridLayout** so labels and text fields align neatly.

📄 activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:text="Student Details"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:layout_marginBottom="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:columnCount="2"
        android:rowCount="3"
        android:alignmentMode="alignMargins"
        android:useDefaultMargins="true">

        <TextView
            android:text="Student Name:"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etName"
            android:hint="Enter Name"
            android:layout_width="200dp"
            android:layout_height="wrap_content"/>

        <TextView
            android:text="Class:"
```

```
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etClass"
            android:hint="Enter Class"
            android:layout_width="200dp"
            android:layout_height="wrap_content"/>

        <TextView
            android:text="Roll No:"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etRoll"
            android:hint="Enter Roll No"
            android:inputType="number"
            android:layout_width="200dp"
            android:layout_height="wrap_content"/>

    </GridLayout>

</LinearLayout>
```

✅ **Proper alignment achieved**

- Labels on the left
- Textboxes aligned on the right

---

3. MainActivity.java

```
package com.example.studentdetails;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

---

4. Execute the Application on Virtual Device

Steps:

1. Open **AVD Manager** (Android Virtual Device)
2. Click **Run** ▶
3. Select the emulator

4. App launches and displays the student form.

Assignment 2: Set A

To create a simple Android application in **Java** to demonstrate the **Activity Life Cycle** and observe the output in **Logcat**.

Activity Life Cycle Methods Used

- `onCreate()`
- `onStart()`
- `onResume()`
- `onPause()`
- `onStop()`
- `onRestart()`
- `onDestroy()`

---

1. Create Android Project

- Open **Android Studio**
- New Project → **Empty Activity**
- App Name: `ActivityLifeCycle`
- Language: **Java**
- Finish

---

2. MainActivity.java (Life Cycle Code)

```
package com.example.activitylifecycle;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d("MyActivity", "onCreate() method called");
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("MyActivity", "onStart() method called");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d("MyActivity", "onResume() method called");
    }
```

```
    @Override
    protected void onPause() {
        super.onPause();
        Log.d("MyActivity", "onPause() method called");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d("MyActivity", "onStop() method called");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d("MyActivity", "onRestart() method called");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("MyActivity", "onDestroy() method called");
    }
}
```

How to Check Logcat Output

Steps:

1. Run the app on **Emulator**
2. Open **Logcat** window (bottom panel)
3. Filter by:
4. `MyActivity`
5. Observe messages while:
    - Launching app
    - Pressing **Home**
    - Reopening app
    - Pressing **Back**
    - Rotating screen

Set A: 2) Create a Simple Application Which Send Hello message from one activity to another with help of Button (Use Intent).

1. Create Android Project

- Open **Android Studio**
- New Project → **Empty Activity**
- App Name: `HelloIntentApp`
- Language: **Java**
- Finish

---

2. First Activity (Sender Activity)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnNext"
        android:text="Next"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

📄 MainActivity.java

```
package com.example.hellointentapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    Button btnNext;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNext = findViewById(R.id.btnSend);

        btnNext.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
            public void onClick(View view) {
      Intent intent = new Intent(MainActivity.this, SecondActivity.class);
                intent.putExtra("message", "Hello from Main Activity.");
                startActivity(intent);
            }
        });
    }
}
```

---

3. Second Activity (Receiver Activity)

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tvMessage"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

■ SecondActivity.java

```
package com.example.hellointentapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.widget.TextView;

public class SecondActivity extends AppCompatActivity {

    TextView tvMessage;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        tvMessage = findViewById(R.id.tvMessage);

        Intent intent = getIntent();
        String msg = intent.getStringExtra("message");

        tvMessage.setText(msg);
    }
}
```

Output

- First screen shows **Next** button
- On clicking button → second screen displays **Hello from Main Activity.**

Set A: 3) Create Simple application to display details of selected list item on Second Activity (use Fragmentation).

- **Activity 1** shows a list
- When you tap an item → **Activity 2** opens
- **Activity 2 hosts a Fragment** that displays the selected item's details

So the data flow is:

```
MainActivity → Intent → DetailActivity → Bundle → DetailFragmen
```

📱 App Structure

```
MainActivity        → shows ListView
DetailActivity      → hosts DetailFragment
DetailFragment      → displays selected item
```

---

1. MainActivity (List of items)

MainActivity.java

```
package com.example.fragmentlistdemo;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listView;
    String[] items = {"Stationary", "Furniture", "Electronics", "Food"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        ArrayAdapter<String> adapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_list_item_1,
                        items);
        listView.setAdapter(adapter);

        listView.setOnItemClickListener(new
AdapterView.OnItemClickListener() {
            @Override
      public void onItemClick(AdapterView<?> parent, View view, int
position, long id) {
```

```
        Intent intent = new Intent(MainActivity.this, DetailActivity.class);
                intent.putExtra("selectedItem", items[position]);
                startActivity(intent);
            }
        });
    }
}
```

---

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
</LinearLayout>
```

---

2. DetailActivity (Hosts Fragment)

DetailActivity.java

```
package com.example.fragmentlistdemo;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class DetailActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_detail);

        if (savedInstanceState == null) {
            DetailFragment fragment = new DetailFragment();
            Bundle bundle = new Bundle();
        bundle.putString("item",getIntent().getStringExtra("selectedItem"));

            fragment.setArguments(bundle);
            getSupportFragmentManager()
                    .beginTransaction()
                    .replace(R.id.fragment_container, fragment)
                    .commit();
        }
    }
}
```

---

activity_detail.xml

```
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

3. DetailFragment (Displays selected item)

DetailFragment.java

```
package com.example.fragmentlistdemo;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class DetailFragment extends Fragment {

    @Nullable
    @Override
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

  View view = inflater.inflate(R.layout.fragment_detail, container, false);

        TextView textView = view.findViewById(R.id.textView);

        if (getArguments() != null) {
            String item = getArguments().getString("item");
            textView.setText("Selected Item: " + item);
        }

        return view;
    }
}
```

fragment_detail.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:id="@+id/textView"
        android:textSize="24sp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

Output

- First screen → List of items
- Click any item → Second Activity opens
- Selected item displayed using **Fragment**

Set B: 1) Create a simple application for Login Screen which contains email, password and login button. On successful login, gives the message **"Login Successful"** with different font color, size and background color on next Activity.

App Flow

1. **LoginActivity**
   - Email
   - Password
   - Login Button
2. **SuccessActivity**
   - Shows **Login Successful** message with custom style

---

1. Create Android Project

- New Project → **Empty view Activity**
- App Name: `LoginApp`
- Language: **Java**

---

2. Login Screen UI

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">

    <TextView
        android:text="Login"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etEmail"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
```

```
        android:layout_marginTop="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnLogin"
        android:text="Login"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

---

3. Login Logic (MainActivity.java)

MainActivity.java

```
package com.example.loginapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText etEmail, etPassword;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etEmail = findViewById(R.id.etEmail);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(v -> {

            String email = etEmail.getText().toString();
            String password = etPassword.getText().toString();

            // Simple validation
            if (email.equals("admin@gmail.com") && password.equals("1234"))
{
      Intent intent = new Intent(MainActivity.this, SuccessActivity.class);
                startActivity(intent);
            } else {
          Toast.makeText(this, "Invalid Login", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

---

4. Success Screen UI (Styled Message)

activity_success.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#C8E6C9"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:text="Login Successful"
        android:textColor="#1B5E20"
        android:textSize="26sp"
        android:textStyle="bold"
        android:padding="20dp"
        android:background="#A5D6A7"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

✓ Different **font color**
✓ Different **font size**
✓ Different **background color**

---

5. SuccessActivity.java

SuccessActivity.java

```
package com.example.loginapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class SuccessActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_success);
    }
}
```

Output

- Login screen with Email & Password
- Correct credentials → Next screen
- **"Login Successful"** shown with styled text and background

Set C: To design Android app screens using **Intents** to accept **customer details for Railway Ticket Booking**, display a **welcome message** on the second activity, and show **all details** on the third activity when the **Submit** button is clicked.

App Flow

FirstActivity → SecondActivity → ThirdActivity

---

1. First Activity

activity_first.xml

```
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:text="Railway Ticket Booking"
            android:textSize="20sp"
            android:textStyle="bold"
            android:layout_gravity="center"
            android:layout_marginBottom="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etAadhar"
            android:hint="Aadhar ID"
            android:inputType="number"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etName"
            android:hint="Name"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <!-- Date of Birth -->
        <TextView
            android:text="Date of Birth"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <DatePicker
            android:id="@+id/dpDob"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etAddress"
            android:hint="Address"
```

```
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <!-- Gender -->
        <TextView
            android:text="Gender"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>

        <RadioGroup
            android:id="@+id/rgGender"
            android:orientation="horizontal"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content">

            <RadioButton
                android:text="Male"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"/>

            <RadioButton
                android:text="Female"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"/>
        </RadioGroup>

        <EditText
            android:id="@+id/etContact"
            android:hint="Contact No"
            android:inputType="phone"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <EditText
            android:id="@+id/etEmail"
            android:hint="Email ID"
            android:inputType="textEmailAddress"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

        <Button
            android:id="@+id/btnNext"
            android:text="Next"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"/>

    </LinearLayout>
</ScrollView>
```

FirstActivity.java

```
package com.example.railwaybooking;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.widget.*;

public class FirstActivity extends AppCompatActivity {

    EditText etAadhar, etName, etAddress, etContact, etEmail;
    DatePicker dpDob;
```

```
    RadioGroup rgGender;
    Button btnNext;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_first);

        etAadhar = findViewById(R.id.etAadhar);
        etName = findViewById(R.id.etName);
        etAddress = findViewById(R.id.etAddress);
        etContact = findViewById(R.id.etContact);
        etEmail = findViewById(R.id.etEmail);
        dpDob = findViewById(R.id.dpDob);
        rgGender = findViewById(R.id.rgGender);
        btnNext = findViewById(R.id.btnNext);

        btnNext.setOnClickListener(v -> {

            int day = dpDob.getDayOfMonth();
            int month = dpDob.getMonth() + 1;
            int year = dpDob.getYear();
            String dob = day + "/" + month + "/" + year;

            RadioButton rb =
findViewById(rgGender.getCheckedRadioButtonId());
            String gender = rb.getText().toString();

            Intent intent = new Intent(this, SecondActivity.class);
            intent.putExtra("aadhar", etAadhar.getText().toString());
            intent.putExtra("name", etName.getText().toString());
            intent.putExtra("dob", dob);
            intent.putExtra("address", etAddress.getText().toString());
            intent.putExtra("gender", gender);
            intent.putExtra("contact", etContact.getText().toString());
            intent.putExtra("email", etEmail.getText().toString());

            startActivity(intent);
        });
    }
}
```

---

2. Second Activity

activity_second.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tvWelcome"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
```

```
    <Button
        android:id="@+id/btnSubmit"
        android:text="Submit"
        android:layout_marginTop="20dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

SecondActivity.java

```
package com.example.railwaybooking;
mport androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.content.Intent;
import android.widget.*;

public class SecondActivity extends AppCompatActivity {

    TextView tvWelcome;
    Button btnSubmit;

    String aadhar, name, dob, address, gender, contact, email;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        tvWelcome = findViewById(R.id.tvWelcome);
        btnSubmit = findViewById(R.id.btnSubmit);

        Intent i = getIntent();
        aadhar = i.getStringExtra("aadhar");
        name = i.getStringExtra("name");
        dob = i.getStringExtra("dob");
        address = i.getStringExtra("address");
        gender = i.getStringExtra("gender");
        contact = i.getStringExtra("contact");
        email = i.getStringExtra("email");

        tvWelcome.setText("Welcome " + name + " to Railway Booking");

        btnSubmit.setOnClickListener(v -> {
            Intent intent = new Intent(this, ThirdActivity.class);
            intent.putExtra("details",
                    "Aadhar ID: " + aadhar +
                    "\nName: " + name +
                    "\nDOB: " + dob +
                    "\nAddress: " + address +
                    "\nGender: " + gender +
                    "\nContact: " + contact +
                    "\nEmail: " + email);
            startActivity(intent);
        });
    }
}
```

---

3. Third Activity

activity_third.xml

```
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/tvDetails"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:textSize="16sp"
    android:padding="16dp"/>
```

ThirdActivity.java

```
package com.example.railwaybooking;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class ThirdActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_third);

        TextView tv = findViewById(R.id.tvDetails);
        tv.setText(getIntent().getStringExtra("details"));
    }
}
```

---

4. AndroidManifest.xml

```
<application ... >
    <activity android:name=".ThirdActivity"/>
    <activity android:name=".SecondActivity"/>
    <activity android:name=".FirstActivity">
        <intent-filter>
            <action android:name="android.intent.action.MAIN"/>
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>
</application>
```

Result

The Android application successfully accepts **customer details**, displays a **welcome message**, and shows **all entered information** on the third activity using **Intents**.